

1. Run interactive job:

```
[ares][plguser@login01 ppss_2026]$ srun -p plgrid-now -N 1 --ntasks-per-node=1 -n 1 -A  
plgbelle2ml26-cpu -t 1:00:00 --pty /bin/bash -l
```

2. Load modules:

```
[ares][plguser@ac0788 ppss_2026]$ module load gcccore/14.3.0 python/3.13.5  
gcccore/14.3.0 loaded.  
zlib/1.3.1-gcccore-14.3.0 loaded.  
binutils/2.44-gcccore-14.3.0 loaded.  
bzip2/1.0.8-gcccore-14.3.0 loaded.  
ncurses/6.5-gcccore-14.3.0 loaded.  
libreadline/8.2-gcccore-14.3.0 loaded.  
libtommath/1.3.0-gcccore-14.3.0 loaded.  
tcl/9.0.1-gcccore-14.3.0 loaded.  
sqlite/3.50.1-gcccore-14.3.0 loaded.  
xz/5.8.1-gcccore-14.3.0 loaded.  
libffi/3.5.1-gcccore-14.3.0 loaded.  
openssl/3 loaded.  
python/3.13.5-gcccore-14.3.0 loaded.
```

3. Install virtual env with python modules **at project space**:

```
[ares][plguser@ac0788 ppss_2026]$ cd $PLG_GROUPS_STORAGE/plggbelle2ml/  
[ares][plguser@ac0788 plggbelle2ml]$ mkdir -p ppss2026venvAres; cd ppss2026venvAres;
```

```
[ares][plguser@ac0788 ppss2026venvAres]$ python3 -m venv jupyter-ppss26-Ares  
[ares][plguser@ac0788 ppss2026venvAres]$ source jupyter-ppss26-Ares/bin/activate  
(jupyter-ppss26-Ares) [ares][plguser@ac0788 ppss2026venvAres]$
```

```
(jupyter-ppss26-Ares) [ares][plguser@ac0788 ppss2026venvAres]$ python3 -m pip  
--no-cache-dir --require-virtualenv install ipykernel uproot numpy matplotlib pandas  
awkward-pandas pyarrow scikit-learn xgboost vector
```

4. Save installed modules with versions:

```
(jupyter-ppss26-Ares) [ares][plguser@ac0788 ppss2026venvAres]$ python3 -m pip  
--require-virtualenv freeze > requirements_jupyter-ppss26-Ares.txt
```

5. Install jupyter kernel.

```
(jupyter-ppss26-Ares) [ares][plguser@ac0788 ppss2026venvAres]$ python3 -m ipykernel  
install --user --name "ppss2026ares" --display-name "3.13.5-ppss2026-Ares" --env  
PYTHONPATH ""  
Installed kernelspec ppss2026ares in  
/net/people/plgrid/plguser/.local/share/jupyter/kernels/ppss2026ares
```